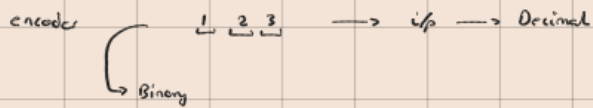
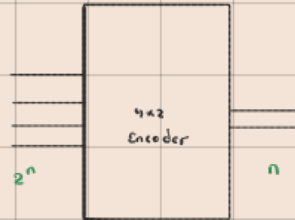


Wed, Apr 8

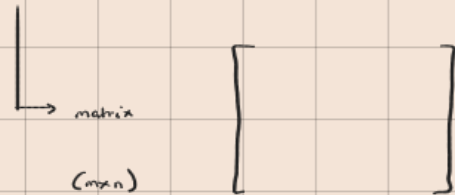
Encoders

- 4x2 Encoders
- 8x3 Encoders
- Decimal to BCD
- Priority Encoders
- Comparators

reverse of Decoders in terms
of functionality



pixel → [:]



ML → Encoder
reduces complexity

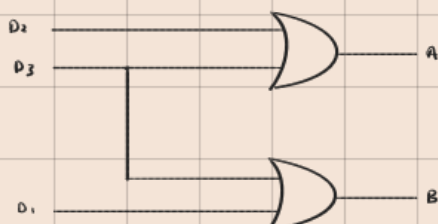
4x2 Encoder



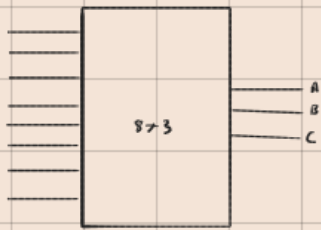
D ₀	D ₁	D ₂	D ₃	A	B
1				0	0
	1			0	1
		1		1	0
			1	1	1
0	0	0	0	x	x

$$A = D_2 + D_3$$

$$B = D_1 + D_3$$



8x3 Encoder

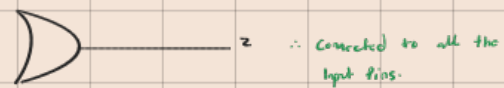
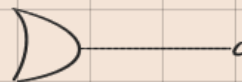
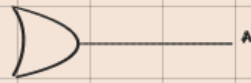


$$A = D_1 + D_5 + D_6 + D_7$$

$$B = D_2 + D_3 + D_6 + D_7$$

$$C = D_0 + D_3 + D_5 + D_7$$

D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	A	B	C
1								0	0	0
	1							0	0	1
		1						0	1	0
			1					0	1	1
				1				1	0	0
					1			1	0	1
						1		1	1	0
							1	1	1	1



Priority Encoder.

scheme (method) $\rightarrow$ higher priority to
higher numbered input pin.

$$D_1 > D_0$$

$$D_2 > D_1$$

$$D_7 > D_6 > \dots > D_0$$

IC 74158 Octal-Binary Priority Encoder.

D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	A	B	C
1								0	0	0
X	1							0	0	1
X	X	1						0	1	0
X	X	X	1					0	1	1
X	X	X	X	1				1	0	0
X	X	X	X	X	1			1	0	1
X	X	X	X	X	X	1		1	1	0
X	X	X	X	X	X	X	1	1	1	1

8) Cascade two 8x3 Encoders to form 16x4

